

## SAFETY DATA SHEET

Creation Date 24-Nov-2010

Revision Date 25-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Osmium(VIII) oxide, 4% aqueous solution

**Cat No. :** 45385

**Synonyms** Osmic acid

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                    |            |
|------------------------------------|------------|
| Acute dermal toxicity              | Category 2 |
| Acute Inhalation Toxicity - Vapors | Category 4 |
| Skin Corrosion/Irritation          | Category 2 |
| Serious Eye Damage/Eye Irritation  | Category 1 |
| Respiratory Sensitization          | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Fatal in contact with skin

Causes skin irritation

Causes serious eye damage  
Harmful if inhaled  
May cause allergy or asthma symptoms or breathing difficulties if inhaled



### Precautionary Statements

#### Prevention

Do not get in eyes, on skin, or on clothing  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
In case of inadequate ventilation wear respiratory protection

#### Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician

#### Skin

Immediately call a POISON CENTER or doctor/physician  
IF ON SKIN: Gently wash with plenty of soap and water  
Remove/Take off immediately all contaminated clothing  
Wash contaminated clothing before reuse  
If skin irritation occurs: Get medical advice/attention

#### Eyes

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

#### Storage

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

None identified

## 3. Composition/Information on Ingredients

| Component        | CAS No     | Weight % |
|------------------|------------|----------|
| Water            | 7732-18-5  | 96       |
| Osmium tetroxide | 20816-12-0 | 4        |

## 4. First-aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Causes eye burns. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Causes severe eye damage. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing                   |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | No information available                                                 |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                 |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                            |

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Keep container tightly closed. Keep refrigerated. Incompatible Materials. Strong oxidizing                                                                                                                                                  |

agents. Halogens. Reducing Agent. Finely powdered metals.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component        | ACGIH TLV                           | OSHA PEL                                                                                                                                                                     | NIOSH                                                                                                                             | Mexico OEL (TWA)                    |
|------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Osmium tetroxide | TWA: 0.0002 ppm<br>STEL: 0.0006 ppm | (Vacated) TWA: 0.0002 ppm<br>(Vacated) TWA: 0.002 mg/m <sup>3</sup><br>(Vacated) STEL: 0.0006 ppm<br>(Vacated) STEL: 0.006 mg/m <sup>3</sup><br>TWA: 0.002 mg/m <sup>3</sup> | IDLH: 1 mg/m <sup>3</sup><br>TWA: 0.0002 ppm<br>TWA: 0.002 mg/m <sup>3</sup><br>STEL: 0.0006 ppm<br>STEL: 0.006 mg/m <sup>3</sup> | TWA: 0.0002 ppm<br>STEL: 0.0006 ppm |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                            |
|----------------------------------------|----------------------------|
| Physical State                         | Liquid                     |
| Appearance                             | Yellow                     |
| Odor                                   | Slight chlorine            |
| Odor Threshold                         | No information available   |
| pH                                     | 6 @ 20°C 25 g/l aq.sol     |
| Melting Point/Range                    | No data available          |
| Boiling Point/Range                    | 100 °C / 212 °F @ 760 mmHg |
| Flash Point                            | No information available   |
| Evaporation Rate                       | No information available   |
| Flammability (solid,gas)               | Not applicable             |
| Flammability or explosive limits       |                            |
| Upper                                  | No data available          |
| Lower                                  | No data available          |
| Vapor Pressure                         | No information available   |
| Vapor Density                          | No information available   |
| Specific Gravity                       | 1.040                      |
| Solubility                             | Soluble                    |
| Partition coefficient; n-octanol/water | No data available          |

|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | No information available |
| Decomposition Temperature | No information available |
| Viscosity                 | No information available |

## 10. Stability and reactivity

|                                  |                                                                           |
|----------------------------------|---------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                |
| Stability                        | Stable under normal conditions.                                           |
| Conditions to Avoid              | Incompatible products.                                                    |
| Incompatible Materials           | Strong oxidizing agents, Halogens, Reducing Agent, Finely powdered metals |
| Hazardous Decomposition Products | None under normal use conditions                                          |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                  |
| Hazardous Reactions              | None under normal processing.                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                     |
|-------------|-------------------------------------|
| Oral LD50   | Category 4. ATE = 300 - 2000 mg/kg. |
| Dermal LD50 | Category 2. ATE = 50 - 200 mg/kg.   |
| Vapor LC50  | Category 4. ATE = 10 - 20 mg/l.     |

#### Component Information

| Component        | LD50 Oral        | LD50 Dermal | LC50 Inhalation           |
|------------------|------------------|-------------|---------------------------|
| Water            | -                | -           | -                         |
| Osmium tetroxide | 15 mg/kg ( Rat ) | Not listed  | LC50 = 40 ppm ( Rat ) 4 h |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to respiratory system and skin CAUSES (SEVERE) EYE BURNS                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component        | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|------------|------------|------------|------------|------------|------------|
| Water            | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Osmium tetroxide | 20816-12-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                      |                           |
|----------------------|---------------------------|
| Reproductive Effects | No information available. |
|----------------------|---------------------------|

|                       |                           |
|-----------------------|---------------------------|
| Developmental Effects | No information available. |
|-----------------------|---------------------------|

|                |                           |
|----------------|---------------------------|
| Teratogenicity | No information available. |
|----------------|---------------------------|

|                          |            |
|--------------------------|------------|
| STOT - single exposure   | None known |
| STOT - repeated exposure | None known |

|                   |                          |
|-------------------|--------------------------|
| Aspiration hazard | No information available |
|-------------------|--------------------------|

|                                            |                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symptoms / effects, both acute and delayed | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component        | log Pow |
|------------------|---------|
| Osmium tetroxide | 0.9     |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3287  
Proper Shipping Name Toxic liquid, inorganic, n.o.s.  
Technical Name (OSMIUM TETROXIDE)  
Hazard Class 6.1  
Packing Group II

### TDG

UN-No UN3287  
Proper Shipping Name Toxic liquid, inorganic, n.o.s.  
Hazard Class 6.1  
Packing Group II

### IATA

UN-No UN3287  
Proper Shipping Name Toxic liquid, inorganic, n.o.s.  
Hazard Class 6.1  
Packing Group II

### IMDG/IMO

UN-No UN3287  
Proper Shipping Name Toxic liquid, inorganic, n.o.s.  
Hazard Class 6.1  
Packing Group II

## 15. Regulatory information

### United States of America Inventory

| Component        | CAS No     | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|------------------|------------|------|--------------------------------------------------|--------------------------------|
| Water            | 7732-18-5  | X    | ACTIVE                                           | -                              |
| Osmium tetroxide | 20816-12-0 | X    | ACTIVE                                           | -                              |

### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component        | CAS No     | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------|------------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Water            | 7732-18-5  | X   | -   | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Osmium tetroxide | 20816-12-0 | X   | -   | 244-058-7 | X     | -    |      | X    | X     | KE-27435 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component        | CAS No     | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|------------------|------------|----------|-------------------------------|---------------------------------|
| Osmium tetroxide | 20816-12-0 | 4        | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and Health Administration

Not applicable

| Component        | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|------------------|----------------------------------|----------------------------|
| Osmium tetroxide | -                                | TQ: 100 lb                 |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component        | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|------------------|--------------------------|-------------------------------------------|-------------------------------|
| Osmium tetroxide | 1000 lb                  | -                                         | 1000 lb<br>454 kg             |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------------|---------------|------------|--------------|----------|--------------|
| Water            | -             | -          | X            | -        | -            |
| Osmium tetroxide | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant Y

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade No information available

**Authorisation/Restrictions according to EU REACH**

| Component        | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water            | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Osmium tetroxide | 20816-12-0 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component        | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water            | 7732-18-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Osmium tetroxide | 20816-12-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component        | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water            | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Osmium tetroxide | 20816-12-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date**  
**Revision Date**  
**Print Date**

24-Nov-2010  
25-Mar-2024  
25-Mar-2024



**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**